

Problem Set 3

Applied Stats/Quant Methods

Matilda Tomatis

Due: November 16, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in **R** using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.
2. Make a scatterplot of the two variables and add the regression line.
3. Save the residuals of the model in a separate object.
4. Write the prediction equation.

Answer 1.1

A bivariate linear regression is estimated with vote share (**voteshare**) as the dependent variable and the difference in log spending (**difflog**) as the explanatory variable. The estimated coefficient $\hat{\beta}_1 = 0.042$ implies that, for each one-unit increase in the spending difference between the incumbent and the challenger (in log units), the incumbent's vote share increases on average by approximately 0.042.

```
1 question_1 <- lm(voteshare ~ difflog, data = inc.sub)
```

Table 1: First bivariate regression model

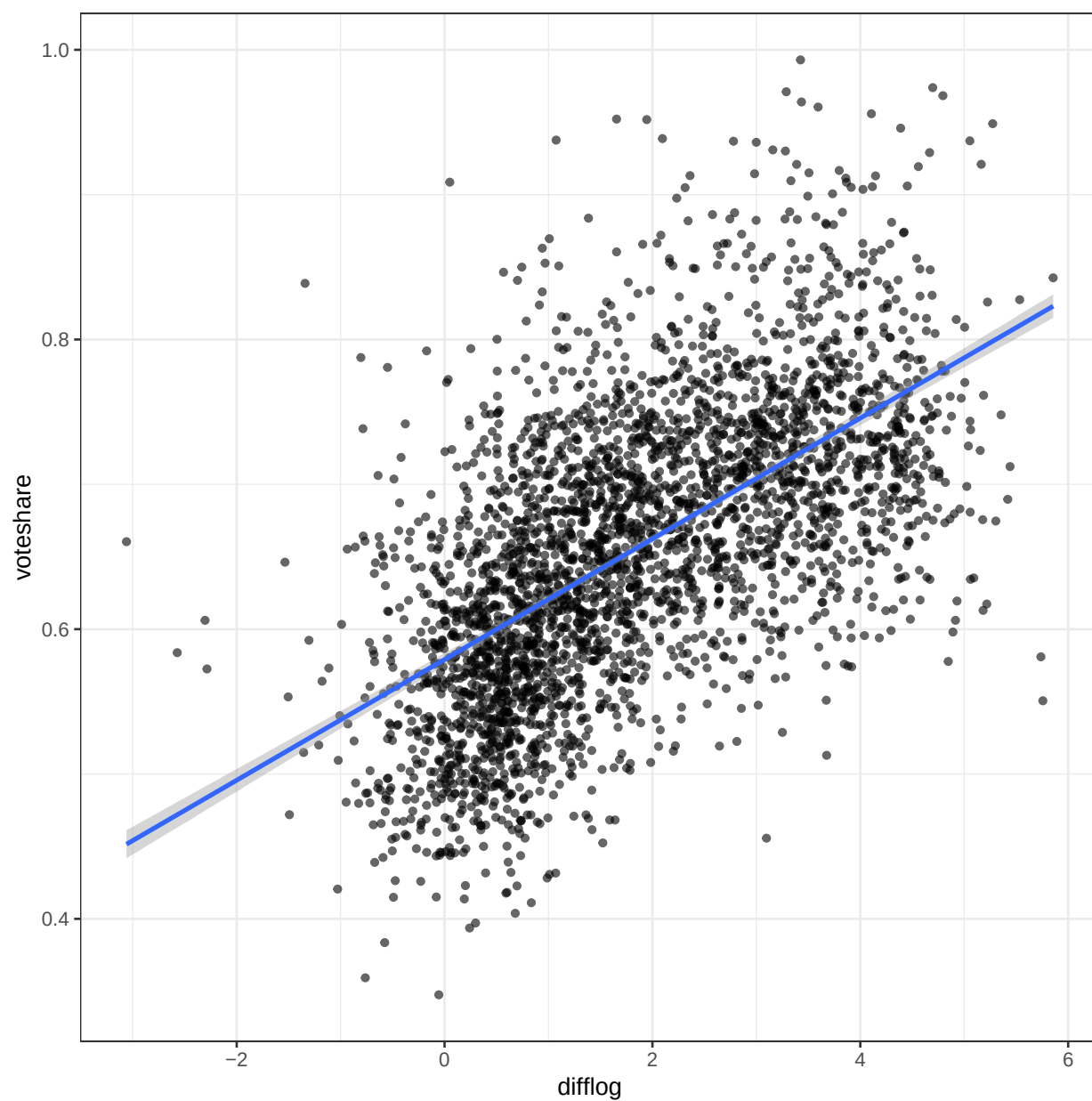
	<i>Dependent variable:</i>
	Incumbent's party overall vote share
Difference in campaign responding	0.042*** (0.001)
Constant	0.579*** (0.002)
Observations	3,193
R ²	0.367
Adjusted R ²	0.367
Residual Std. Error	0.079 (df = 3191)
F Statistic	1,852.791*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Answer 1.2

Follows the code to obtain a scatterplot having **voteshare** on the y-axis and **difflog** on the x-axis, added is their fitted regression line with its shaded confidence interval.

```
1 q1_scatter <- ggplot(inc.sub, aes(difflog, voteshare)) +  
2   geom_point(size = 1.25, alpha = 0.60) +  
3   geom_smooth(method = "lm",  
4               formula = y ~ x) +  
5   theme_bw()
```

Figure 1: Scatterplot of incumbent vote share against spending difference with fitted regression line



Answer 1.3

The residuals obtained from regressing `voteshare` on `difflog` are stored in a separate object.

```
1 q1_residuals <- question_1$residuals
```

Answer 1.4

The prediction equation considers the estimated coefficient to predict unknown `voteshare` values given an appropriate `difflog` Xi.

$$\widehat{\text{voteshare}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog}_i$$

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.
2. Make a scatterplot of the two variables and add the regression line.
3. Save the residuals of the model in a separate object.
4. Write the prediction equation.

Answer 2.1

A bivariate linear regression is estimated with incumbent's party presidential vote share (`presvote`) as the dependent variable and the difference in log spending (`difflog`) as the explanatory variable. The estimated coefficient $\hat{\beta}_1 = 0.023837$ implies that, on average, for each one-unit increase in the spending difference between the incumbent and the challenger (in log units), the presidential candidate of the incumbent's party is expected to have an increase of approximately 2.4 percentage points.

Table 2: Second bivariate regression model

	<i>Dependent variable:</i>
	Presidential candidate's vote share
Difference in campaign responding	0.024*** (0.001)
Constant	0.508*** (0.003)
Observations	3,193
R ²	0.088
Adjusted R ²	0.088
Residual Std. Error	0.110 (df = 3191)
F Statistic	307.715*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

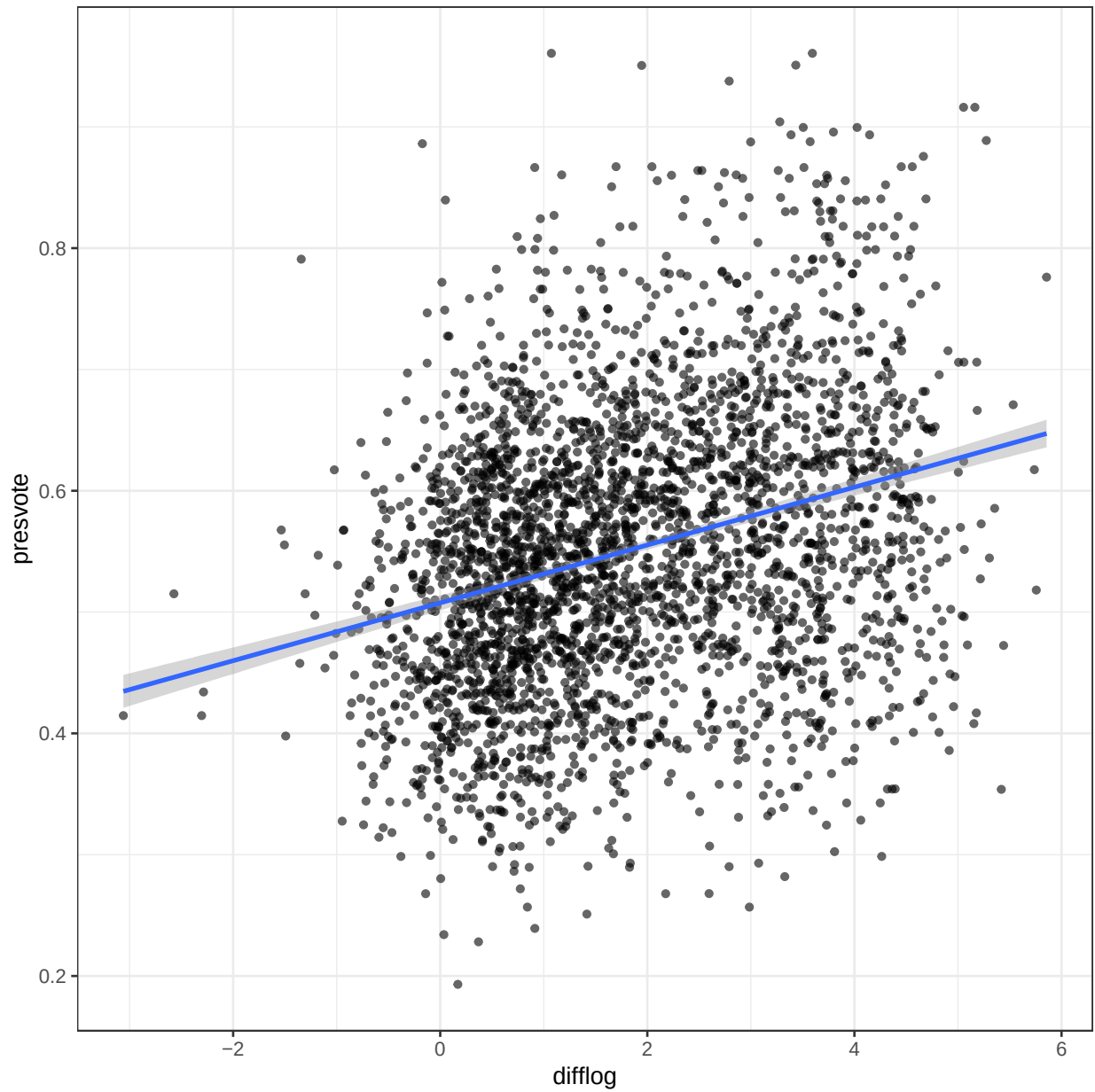
```
1 question_2 <- lm(presvote ~ difflog, data = inc.sub)
```

Answer 2.2

Repeating the same procedure from before we obtain a new scatterplot, this second time **presvote** lays on the y-axis and the regression line is obtained by regressing it on **difflog**.

```
1 q2_scatter <- ggplot(inc.sub, aes(difflog, presvote)) +
2   geom_point(size = 1.25, alpha = 0.60) +
3   geom_smooth(method = "lm",
4     formula = y ~ x) +
5   theme_bw()
```

Figure 2: Scatterplot of incumbent's party presidential candidate's vote share against spending difference, with fitted regression line



Answer 2.3

The residuals obtained from the second simple linear regression are stored in a separate object.

```
1 question_3 <- lm(voteshare ~ presvote, data = inc.sub)
```

Answer 2.4

The prediction equation considers the estimated coefficient to predict unknown **presvote** values given an appropriate **difflog** Xi.

$$\widehat{\text{presvote}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog}_i$$

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is **voteshare** and the explanatory variable is **presvote**.
2. Make a scatterplot of the two variables and add the regression line.
3. Write the prediction equation.

Answer 3.1

For the last estimation of the β_1 and β_0 coefficients using data from the **incumbent_subset** dataset, a bivariate regression with **voteshare** as the dependent variable and **presvote** as the explanatory one. The estimated coefficient $\hat{\beta}_1 = 0.388$ suggests that, on average, when the incumbent party's presidential candidate performs better by one unit in vote share, the incumbent's own vote share increases by approximately 0.388 units

```
1 question_3 <- lm(voteshare ~ presvote, data = inc.sub)
```

Table 3: Third bivariate regression model

	<i>Dependent variable:</i>
	Incumbent's party overall vote share
Presidential candidate's vote share	0.388*** (0.013)
Constant	0.441*** (0.008)
Observations	3,193
R ²	0.206
Adjusted R ²	0.206
Residual Std. Error	0.088 (df = 3191)
F Statistic	826.950*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

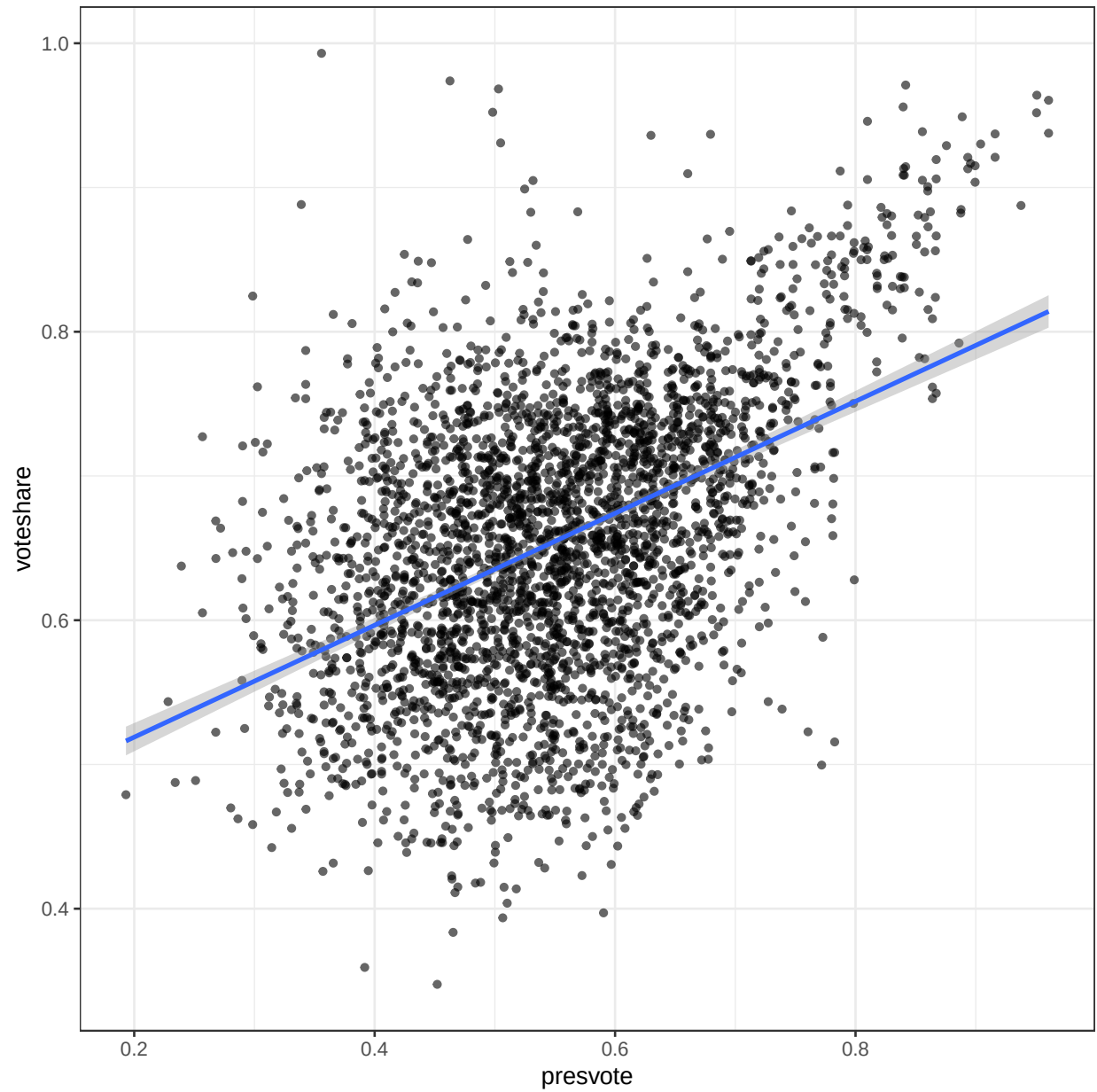
Answer 3.2

Once again, using the same methods, a scatterplot of our two variables of interest is made and their simple linear regression line is added.

```

1 q3_scatter <- ggplot(inc.sub, aes(presvote, voteshare)) +
2   geom_point(size = 1.25, alpha = 0.60) +
3   geom_smooth(method = "lm",
4     formula = y ~ x) +
5   theme_bw()
```

Figure 3: Scatterplot of incumbent's party overall vote share against incumbent's party presidential candidate's vote share, with fitted regression line



Answer 3.3

It's presented is the prediction equation for new incumbent's party vote share given a in-

incumbent's party presidential candidate's vote share.

$$\widehat{\text{voteshare}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{presvote}_i$$

Question 4

The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.
2. Make a scatterplot of the two residuals and add the regression line.
3. Write the prediction equation.

Answer 4.1

Using the vectors created when storing the linear regression residuals from Q1 and Q2, a bivariate linear regression is performed. The estimated $\hat{\beta}_1$ coefficient provides an indication of the magnitude and direction of the average linear relationship between the incumbent party's overall vote share and the incumbent party's presidential candidate's vote share, after controlling for the difference in campaign spending between the incumbent and the challenger. The significance of the estimate suggests that it would be appropriate to control for both **difflog** and **presvote** when estimating the **voteshare** equation.

```
1 question_4 <- lm(q1_residuals ~ q2_residuals)
```

Table 4: Regressing Q1 residuals on Q2 residuals

	<i>Dependent variable:</i>
	Question 1 residuals
Question 2 residuals	0.257*** (0.012)
Constant	-0.000 (0.001)
Observations	3,193
R ²	0.130
Adjusted R ²	0.130
Residual Std. Error	0.073 (df = 3191)
F Statistic	476.975*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Answer 4.2

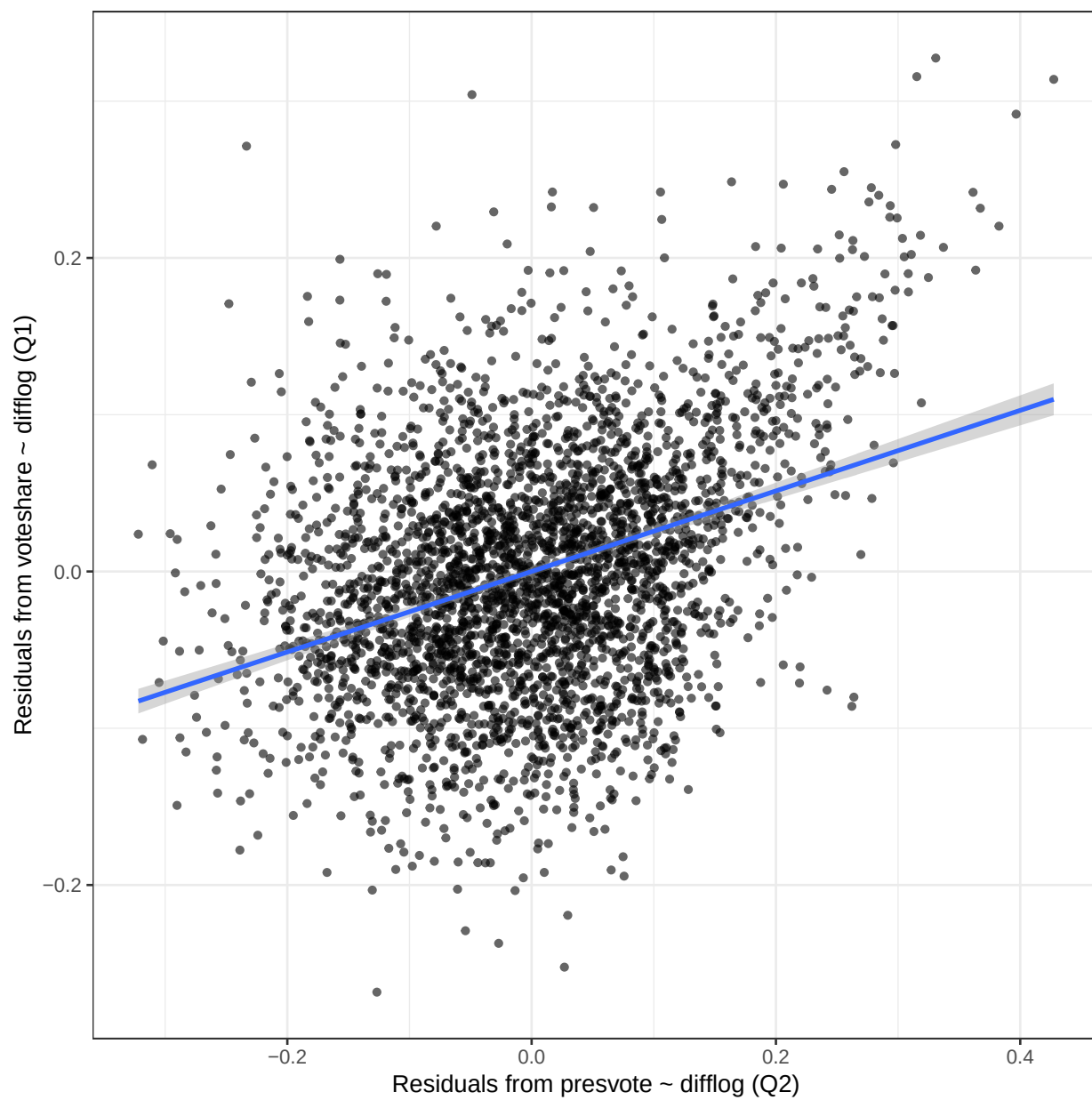
In order to input the needed data for the creation of a scatterplot a dataframe is created with Q1 and Q2 residuals vectors. Afterwards the fitted line is added.

```

1 q4_scatter <- ggplot(data = data.frame(q1_residuals, q2_residuals),
2                           aes(x = q2_residuals, y = q1_residuals)) +
3   geom_point(size = 1.25, alpha = 0.60) +
4   geom_smooth(method = "lm",
5               formula = y ~ x) +
6   labs(
7     x = "Residuals from presvote ~ difflog (Q2)",
8     y = "Residuals from voteshare ~ difflog (Q1)") +
9   theme_bw()

```

Figure 4: Scatterplot of Q1 residuals against Q2 residuals, with fitted regression line



Answer 4.3

Here follows the prediction equation for Q1 residuals given Q2 residual values.

$$\widehat{\text{q1_residuals}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{q2_residuals}_i$$

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.
2. Write the prediction equation.
3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

Answer 5.1

Differently from before, a multivariate linear regression is now run. The dependent variable is `voteshare`, while both `presvote` and `difflog` are used as explanatory variables. This fifth regression allows for controlling differences in spending between the incumbent's and challenger's parties when estimating the effect of the presidential candidate's vote share on the overall incumbent party's vote share, and vice versa. In simpler terms, the coefficients obtained from this regression can be interpreted as the effect of each explanatory variable on the dependent variable, holding the other independent variable constant.

```
1 question_5 <- lm(voteshare ~ presvote + difflog, data = inc.sub)
```

Table 5: Multivariate regression model

	<i>Dependent variable:</i>
	Incumbent's party overall vote share
Presidential candidate's vote share	0.257*** (0.012)
Difference in campaign spending	0.036*** (0.001)
Constant	0.449*** (0.006)
Observations	3,193
R ²	0.450
Adjusted R ²	0.449
Residual Std. Error	0.073 (df = 3190)
F Statistic	1,302.947*** (df = 2; 3190)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Answer 5.2

The prediction equation of the multivariate regression estimated above is the following:

$$\widehat{\text{voteshare}_i} = \hat{\beta}_0 + \hat{\beta}_1 \text{presvote}_i + \hat{\beta}_2 \text{difflog}_i$$

Answer 5.3

The $\hat{\beta}_1$ coefficients are identical in the bivariate residuals regression estimated in Question 4 and in the multivariate regression run in Question 5. This occurs because the residuals obtained in Questions 1 and 2 represent the remaining variation in **voteshare** and **presvote** after controlling for **difflog**. Regressing one set of residuals on the other therefore captures the extent to which the unexplained variation in one variable is associated with that in the other. This is the same underlying mechanism as in a multivariate regression, where each explanatory variable's coefficient is estimated while holding the others constant, thereby

controlling for their effects. The $\hat{\beta}_1 = 0.257$ coefficient is interpretable as the expected average change in incumbent's party overall vote share given a unit change in incumbent's party presidential candidate's vote share, keeping the difference in spending between parties constant.